

Solutions Of Problems on CH 29

2. Determine the initial direction of the deflection of charged particles as they enter the magnetic fields shown in Figure P29.2.

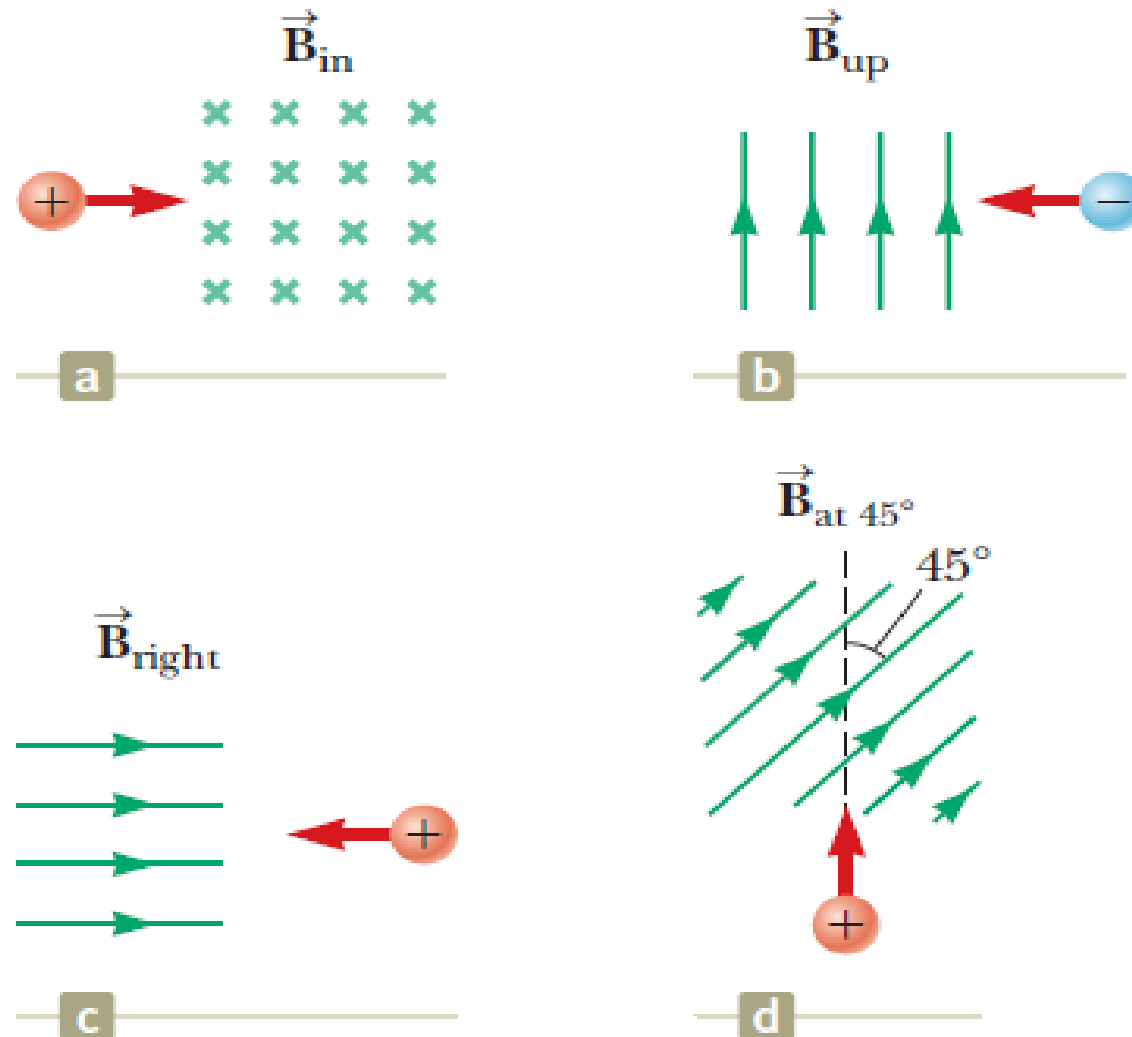


Figure P29.2

- (a) up
- (b) out of the page, since the charge is negative.
- (c) no deflection
- (d) into the page

3. Find the direction of the magnetic field acting on a positively charged particle moving in the various situations shown in Figure P29.3 if the direction of the magnetic force acting on it is as indicated.

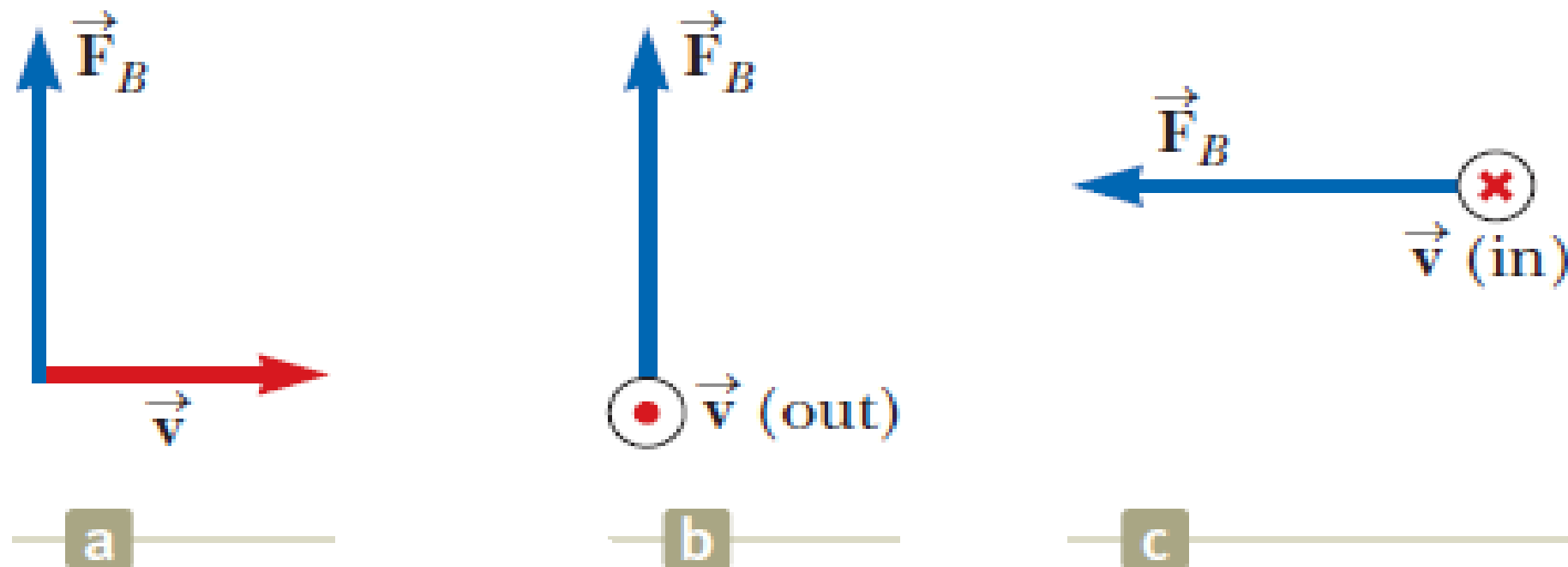


Figure P29.3

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- (a) into the page (b) toward the right
- (c) toward the bottom of the page

8. A proton moves with a velocity of $\vec{v} = (2\hat{i} - 4\hat{j} + \hat{k})$ m/s in a region in which the magnetic field is $\vec{B} = (\hat{i} + 2\hat{j} - \hat{k})$ T. What is the magnitude of the magnetic force this particle experiences?



$$\vec{\mathbf{F}}_B = q\vec{\mathbf{v}} \times \vec{\mathbf{B}} = (1.60 \times 10^{-19} \text{ C}) \left[(2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + \hat{\mathbf{k}})(\text{m/s}) \times (\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}) \text{ T} \right]$$

$$\vec{\mathbf{F}}_B = (1.60 \times 10^{-19} \text{ N}) \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -4 & 1 \\ 1 & 2 & -1 \end{vmatrix}$$

$$\vec{\mathbf{F}}_{B,x} = (1.60 \times 10^{-19} \text{ N}) \left[(-4)(-1) - (1)(2) \right] \hat{\mathbf{i}}$$

$$\vec{\mathbf{F}}_{B,y} = (1.60 \times 10^{-19} \text{ N}) \left[(1)(1) - (2)(-1) \right] \hat{\mathbf{j}}$$

$$\vec{\mathbf{F}}_{B,z} = (1.60 \times 10^{-19} \text{ N}) \left[(2)(2) - (1)(-4) \right] \hat{\mathbf{k}}$$

$$\begin{aligned} \vec{\mathbf{F}}_B &= (1.60 \times 10^{-19} \text{ N})(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 8\hat{\mathbf{k}}) \\ &= (3.20\hat{\mathbf{i}} + 4.80\hat{\mathbf{j}} + 12.8\hat{\mathbf{k}}) \times 10^{-19} \text{ N} \end{aligned}$$

$$|\vec{\mathbf{F}}_B| = \left(\sqrt{3.20^2 + 4.80^2 + 12.8^2} \right) \times 10^{-19} \text{ N} = \boxed{13.2 \times 10^{-19} \text{ N}}$$

13. An electron moves in a circular path perpendicular to a uniform magnetic field with a magnitude of 2.00 mT. If the speed of the electron is 1.50×10^7 m/s, determine (a) the radius of the circular path and (b) the time interval required to complete one revolution.

$$(a) \quad r = \frac{m_e v}{eB} = \frac{(9.11 \times 10^{-31} \text{ kg})(1.50 \times 10^7 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(2.00 \times 10^{-3} \text{ T})}$$
$$= 0.0427 \text{ m} = \boxed{4.27 \text{ cm}}$$

$$(b) \quad T = \frac{\text{distance traveled}}{\text{constant speed}} = \frac{2\pi r}{v} = \frac{2\pi(0.0427 \text{ m})}{1.50 \times 10^7 \text{ m/s}} = \boxed{1.79 \times 10^{-8} \text{ s}}$$

19. Review. An electron moves in a circular path perpendicular to a constant magnetic field of magnitude 1.00 mT. The angular momentum of the electron about the center of the circle is $4.00 \times 10^{-25} \text{ kg} \cdot \text{m}^2/\text{s}$. Determine (a) the radius of the circular path and (b) the speed of the electron.

(a) We begin with $qvB = \frac{mv^2}{R}$, or $qRB = mv$.

But, $L = mvR = qR^2B$.

Therefore,

$$R = \sqrt{\frac{L}{qB}} = \sqrt{\frac{4.00 \times 10^{-25} \text{ J} \cdot \text{s}}{(1.60 \times 10^{-19} \text{ C})(1.00 \times 10^{-3} \text{ T})}} = 0.0500 \text{ m}$$
$$= \boxed{5.00 \text{ cm}}$$

(b) Thus,

$$v = \frac{L}{mR} = \frac{4.00 \times 10^{-25} \text{ J} \cdot \text{s}}{(9.11 \times 10^{-31} \text{ kg})(0.0500 \text{ m})} = \boxed{8.78 \times 10^6 \text{ m/s}}$$

33. A conductor carrying a current $I = 15.0 \text{ A}$ is directed along the positive x axis and perpendicular to a uniform magnetic field. A magnetic force per unit length of 0.120 N/m acts on the conductor in the negative y direction. Determine (a) the magnitude and (b) the direction of the magnetic field in the region through which the current passes.

(a) From $F = BIL \sin \theta$, the magnetic field is

$$B = \frac{F/L}{I \sin \theta} = \frac{0.120 \text{ N/m}}{(15.0 \text{ A}) \sin 90^\circ} = \boxed{8.00 \times 10^{-3} \text{ T}}$$

(b) The magnetic field must be in the $+z$ direction to produce a force in the $-y$ direction when the current is in the $+x$ direction.

35. A wire carries a steady current of 2.40 A. A straight section of the wire is 0.750 m long and lies along the x axis within a uniform magnetic field, $\vec{\mathbf{B}} = 1.60\hat{\mathbf{k}}$ T. If the current is in the positive x direction, what is the magnetic force on the section of wire?

$$\vec{\mathbf{F}}_B = I\vec{\ell} \times \vec{\mathbf{B}} = (2.40 \text{ A})(0.750 \text{ m})\hat{\mathbf{i}} \times (1.60 \text{ T})\hat{\mathbf{k}} = \boxed{(-2.88\hat{\mathbf{j}}) \text{ N}}$$

39. A wire having a mass per unit length of 0.500 g/cm carries a 2.00-A current horizontally to the south. What are (a) the direction and (b) the magnitude of the minimum magnetic field needed to lift this wire vertically upward?

(a) east

(b) $F_B = ILB \sin \theta$ with $F_B = F_g = mg$

$$mg = ILB \sin \theta \quad \text{so} \quad \frac{m}{L} g = IB \sin \theta \quad \rightarrow \quad B = \frac{m}{L} \frac{g}{I \sin \theta}$$

$$B = \frac{m}{L} \frac{g}{I \sin \theta} = \left(\frac{0.500 \times 10^{-3} \text{ kg}}{1.00 \times 10^{-2} \text{ m}} \right) \left(\frac{9.80 \text{ m/s}^2}{(2.00 \text{ A}) \sin 90.0^\circ} \right) = \boxed{0.245 \text{ T}}$$